LINUX INTERVIEW QUESTIONS AND ANSWERS

1. What is Linux and how is it different from Windows?

Linux is an open-source operating system based on the Linux kernel. Windows is a proprietary operating system developed by Microsoft. Linux is highly customizable and commonly used for servers, cloud, and DevOps, while Windows is widely used for desktops and enterprise applications.

2. What is the Linux kernel?

The Linux kernel is the core component of the Linux operating system. It manages CPU, memory, processes, devices, networking, and communication between hardware and software. Users interact with the kernel through commands and applications.

3. Explain the Linux file system hierarchy.

Linux organizes files in a tree-like structure starting from /, called the root directory. Important directories include /etc for configuration, /var for logs, /home for users, /tmp for temporary files, and /usr for applications and utilities.

4. Difference between absolute and relative paths?

An absolute path starts from the root /, for example /home/mayur/file.txt. A relative path starts from the current directory, for example docs/file.txt. Absolute paths always identify the same location regardless of the current directory.

5. What are cat, less, head, and tail used for?

cat displays the complete contents of a file. less allows you to read a large file page by page. head shows the beginning of a file, while tail shows the end, commonly used for checking logs.

6. What are Vim modes?

The main Vim modes are Normal, Insert, Command-line, and Visual mode. Normal mode is used for navigation and commands. Insert mode is used for typing, Command-line mode executes commands such as :wq, and Visual mode selects text.

7. How do you check system information?

uname -a displays kernel and system information. hostname shows the system hostname. df -h checks disk usage and free -h checks memory usage.

ADVANCED QUESTIONS

1. Explain Linux permissions and rwx.

Linux permissions are divided into user, group, and others. r means read, w means write, and x means execute. For a file, rwxr-xr-- means the owner has all permissions, the group has read/execute, and others have read only.

2. Difference between symbolic link and hard link.

A symbolic link is a shortcut that points to another file path/inode. A hard link points directly to the same inode as the original file. A symbolic link can point to directories and filesystems, while hard links generally cannot cross filesystems.

3. Difference between chown and chgrp.

chown changes the owner of a file or directory. chgrp changes the group ownership. For example, chown user file changes the owner, while chgrp developers file changes the group.

4. How does SUDO work

sudo allows an authorized user to execute commands with another user's privileges, usually root. It checks permissions defined mainly in /etc/sudoers and related configuration. It provides controlled administrative access without giving users the root password.

5. What is the difference between the hard and soft links

A hard link points directly to the same inode as the original file. A soft link stores the path of another file and works like a shortcut. If the original file is deleted, a hard link can still access the data, but a soft link becomes broken.

6. Explain /etc/passwd and /etc/shadow.

/etc/passwd contains basic information about Linux users such as username, UID, GID, home directory, and login shell. /etc/shadow stores password hashes and password-aging information. /etc/shadow is restricted because it contains sensitive authentication data.

7. How do you manage users and groups?

Users can be created with useradd or adduser and deleted with userdel. Groups can be created using groupadd. Commands such as usermod, passwd, and gpasswd are used to modify users, passwords, and group membership.

8. Explain the UMASK

umask defines which permissions should be removed from newly created files and directories. A common value is 022. With 022, new files generally get 644 and directories generally get 755.

9. What happens when a cron job fails?

A failed cron job does not normally stop the cron service itself. The command may return an error, and depending on configuration, output/error may be mailed to the user. You should check cron logs, command paths, permissions, environment variables, and manually test the command.

10. How do you debug cron?

First check whether the cron service is running using systemctl status cron or systemctl status crond. Check the cron logs using journalctl or system log files. Also use absolute paths and redirect output/error to a log file for troubleshooting.

11. Difference between tar, gzip, bzip2, and xz.

tar is primarily an archiving tool; it combines multiple files into one archive. gzip, bzip2, and xz are compression tools. Common combinations are tar.gz, tar.bz2, and tar.xz.

12. How does find differ from grep?

find searches for files and directories based on name, size, time, permissions, etc. grep searches for text patterns inside files. For example, find /var -name "*.log" searches for log files, while grep "error" app.log searches for the word error.

13. Explain ps, top, kill, and pkill.

ps displays a snapshot of running processes. top provides real-time process and resource monitoring. kill sends a signal to a process using its PID, while pkill can terminate processes based on their name or other attributes.

14. What is nice and renice?

nice starts a process with a specific CPU scheduling priority. renice changes the nice value of an already-running process. A higher nice value generally means lower CPU priority.

15. Difference between process and thread.

A process is an independent running program with its own memory space. A thread is a smaller execution unit inside a process and shares the process's resources. Multiple threads can exist within one process.

16. Explain ip vs ifconfig.

ip is the modern Linux networking utility used to manage interfaces, addresses, routes, and more. ifconfig is an older utility and may not be installed by default on modern distributions. Today, ip is preferred.

17. How do you configure a static IP using nmcli?

nmcli is used to manage NetworkManager connections. You can configure a static address using commands such as nmcli con mod <connection> ipv4.addresses .... Then configure gateway/DNS and use nmcli con up <connection> to activate it.

18. How do you troubleshoot DNS?

First check connectivity using ping or ip route. Then check DNS configuration using /etc/resolv.conf and test resolution with dig, nslookup, or getent hosts. Also verify whether the DNS server itself is reachable.

19. How do you monitor disk usage?

df -h shows filesystem-level disk usage. du -sh shows the size of a directory. Commands such as du, find, and sort can help identify which directories or files are consuming the most space.

20. How do you identify high CPU/memory processes?

Use top or htop for real-time monitoring. ps aux --sort=-%cpu can identify high CPU processes, while ps aux --sort=-%mem identifies high memory processes. Then investigate the process before terminating it.

SCENARIO-BASED QUESTIONS

21. User cannot access a file due to permissions. Diagnose.

First use ls -l filename to check permissions and ownership. Then use id username to check the user's groups. Depending on the requirement, use chmod, chown, or chgrp to correct the permissions.

22. Developer deleted a file accidentally. Recovery options?

First stop processes that may overwrite the disk if recovery is important. Check backups, snapshots, version control, or application-specific recovery options. On Linux, recovery from an already-deleted file is not guaranteed, so backups are the safest solution.

23. Cron backup didn't run overnight.

Check whether the cron service is running and whether the job exists using crontab -l. Check cron logs and manually execute the backup command. Verify absolute paths, permissions, environment variables, disk space, and destination availability.

24. Disk is 100% full on /var

Run df -h and then investigate with du -sh /var/*. Check large log files, package caches, temporary files, and deleted files still held open by processes. Clean only files that are safe to remove.

25. SSH login suddenly fails.

First check network connectivity and whether port 22 is reachable. On the server, check sshd status and logs using journalctl. Also verify firewall rules, user permissions, SSH configuration, account lockout, and authentication settings.

26. Server unreachable after IP change.

Check the current IP using ip addr and routing using ip route. Verify the network configuration, gateway, firewall, and DNS if a hostname is being used. Make sure clients are connecting to the new IP

27. Application reports Permission denied.

Check the application's user, file ownership, and permissions using ls -l. Also check parent directory permissions because the application needs execute permission to traverse directories. On SELinux systems, also check SELinux context and audit logs.

28. User forgot password.

An administrator can reset the password using passwd username. Before doing so, verify the user's identity and authorization. Also check whether the account is locked or expired.

29. Need a shared directory for a team.

Create a directory and assign it to a dedicated group. Give the group appropriate permissions and add team members to that group. For collaboration, set the directory's group ownership and consider the setgid permission so new files inherit the directory's group.

30. Log file grows too large

Identify the application generating the log and check whether logrotate is configured. Configure rotation based on size/time and optionally compression. Avoid blindly deleting active logs because applications may continue holding deleted files open.

31. High CPU usage on server.

Use top or ps to identify the process consuming CPU. Check application logs and determine whether the behavior is expected or abnormal. If necessary, restart or stop the process according to operational procedures.

32. High memory usage.

Use free -h, top, or ps to identify memory consumption. Determine whether memory is being used by applications, cache, or a possible memory leak. Check swap usage and application logs before taking corrective action.

33. Find files modified in last 24 hours.

Use the find command with -mtime or -mmin. For example: find /path -type f -mtime -1. This searches for regular files modified within approximately the last 24 hours.

34. Archive and compress logs older than 30 days.

First identify the files using find and verify the result. Then use tar with a suitable file list or use find with an appropriate archiving workflow. Always verify the archive before deleting the original logs.

35. Delete files older than 90 days safely.

First perform a dry run: find /path -type f -mtime +90. Review the files carefully before deleting. Once confirmed, use -delete or another controlled deletion method.

36. Service starts manually but not after reboot.

Check whether the service is enabled using systemctl is-enabled service. Review systemctl status and journalctl -u service. Common causes include incorrect dependencies, environment variables, paths, permissions, or startup ordering.

37. Identify process listening on port 8080.

Use ss -lntp | grep :8080 to identify the listening process. You can also use lsof -i :8080. The output normally provides the PID and process name.

38. Recover from wrong chmod -R.

First stop making further permission changes. Determine the correct permissions from backups, another identical server, package defaults, or version-controlled deployment configuration. Avoid blindly running another recursive chmod, because different files require different permissions.

39. Copy 100 GB data preserving permissions.

Use rsync because it efficiently handles large data transfers and can preserve attributes. For example, rsync -a source/ destination/. Verify available disk space and use additional options such as -v or --progress when appropriate.

40. Server time mismatch affects cron.

Check the current time, timezone, and synchronization status using date, timedatectl, and related tools. Configure the correct timezone and enable NTP/time synchronization. Then verify that cron interprets the schedule as expected.

41. Investigate intermittent network connectivity.

Check interface errors, packet loss, routing, DNS, firewall rules, and network statistics. Use tools such as ping, ip, ss, mtr, and tcpdump where appropriate. Compare failures across time and different endpoints.

42. User cannot use sudo.

Check whether the user belongs to the appropriate administrative group or has a sudoers entry. Use sudo -l and inspect /etc/sudoers or /etc/sudoers.d/. Always edit sudo configuration using visudo.

43. File ownership changed unexpectedly.

Check the file's current owner and group using ls -l. Review scripts, deployment tools, cron jobs, containers, or users that could have modified it. System and audit logs may help identify the change.

44. Need password expiry policy.

Linux supports password-aging policies through tools such as chage. You can configure maximum password age, minimum age, warning period, and account expiration. Password policies improve security by limiting the lifetime of credentials.

45. Application cannot resolve hostname.

Test the hostname using getent hosts, nslookup, or dig. Check /etc/resolv.conf, DNS server availability, search domains, and application-specific DNS configuration. Also verify that the hostname actually has a valid DNS record.

46. Background job stopped unexpectedly.

Identify the process and check whether it was killed, crashed, or terminated by the shell. Check ps, journalctl, application logs, and system resource usage. For long-running jobs, tools such as systemd, tmux, or screen may be more suitable.

47. Compare hard vs soft link failure scenario.

If the original file is deleted, a hard link can still access the underlying data because it references the same inode. A soft link stores the original path, so it becomes a broken link. Therefore, hard links are more resilient to deletion of the original filename.

48. Vim swap file appears after crash.

A Vim swap file is created to protect unsaved work and help recover from crashes. If Vim reports a swap file, first determine whether another Vim session is using it. If not, you can recover the file using Vim's recovery option and remove the stale swap file afterward.

49. Production server is slow after deployment. Troubleshoot.

First check CPU, memory, disk, network, and running processes using top, free, df, and ss. Compare the current application version/configuration with the previous version. Check application logs and metrics, and roll back if the deployment is confirmed to be the cause.

50. What is an inode in Linux?

An inode is a data structure that stores metadata about a file, such as its permissions, owner, size, timestamps, and pointers to data blocks. The inode does not normally store the filename itself. Each file has an inode number within its filesystem.
